

TABLE I
BUTTONS AND THEIR FUNCTIONS

Remote control button	Function (Action)
Power	Turn on RGB bulb
0	Turn off RGB bulb
1	Increase red colour intensity
2	Decrease red colour intensity
3	Increase green colour intensity
4	Decrease green colour intensity
5	Increase blue colour intensity
6	Decrease blue colour intensity

TABLE II
CODE FOR DIFFERENT BUTTONS

Remote control button	Code
Power	18615
0	-8161
1	20655
2	-10201
3	-2041
4	12495
5	-20401
6	28815

it to Arduino microcontroller (MCU). Connect IR sensor as shown in the circuit diagram. Point the remote control towards IR sensor and press the corresponding button, as per the details given in Table I.

Open serial monitor of Arduino to observe the code of each button being pressed in the form of numbers as listed in Table II. Note down the code for each button.

These codes are used in Arduino sketch to perform actions as per the details given in Table I. You can use any IR remote after changing the remote button codes in Arduino sketch (program).

Software

Circuit operation is done using the software program loaded in the internal memory of Arduino Uno R3. The program is written in Arduino programming language. Arduino IDE 1.6.4 is used to compile and upload the program.

PARTS LIST

Semiconductors:

Board1 - Arduino Uno board
TSOP1738 - IR receiver module
RGB1 - Common-anode RGB LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1 - 100-ohm

Miscellaneous:

BATT.1 - 9V battery with connector
CON1 - 3-pin connector for TSOP1738
- Jumper wires
- 7-pin connector for board1
IR remote (Vire)

ATmega328P on Arduino Uno comes with a pre-programmed boot loader. It allows you to upload a new code to it without using an external hardware programmer.

The program/sketch (rgb_vire_remote.ino) is the heart of the system and carries out all major functions. In this project, IRremote.h header files are required for programming. You may download the same from GitHub website (<https://github.com/z3t0/Arduino-IRremote>).

Connect Arduino to the PC, and select the correct COM port in Arduino IDE. Compile the program/sketch. Select the correct board from Tools→Board menu in Arduino IDE. Upload the sketch to Arduino's MCU.

Construction and testing

An actual-size PCB layout of the IR remote-controlled RGB bulb is shown in Fig. 5 and its components layout in Fig. 6. Assemble the components on the PCB as per the circuit diagram. Mount RGB1 on the PCB.

Take an incandescent bulb and carefully remove the base including contacts. This is done because you only need the glass envelope in the project. Cover RGB1 with the glass envelope of the bulb as shown in the prototype.

When Power button on the remote is pressed, it turns on RGB1 bulb. Power and 0 buttons on the remote

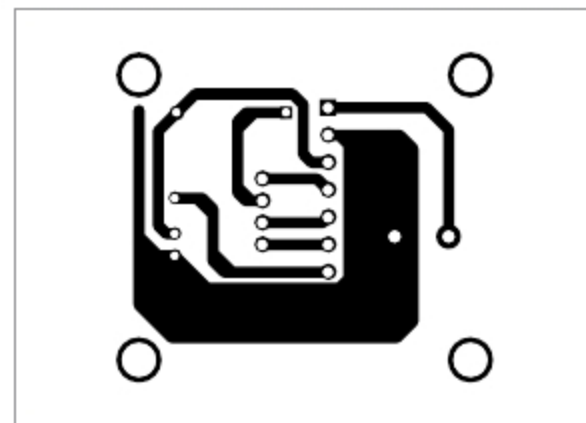


Fig. 5: Actual-size PCB layout of IR remote-controlled RGB bulb

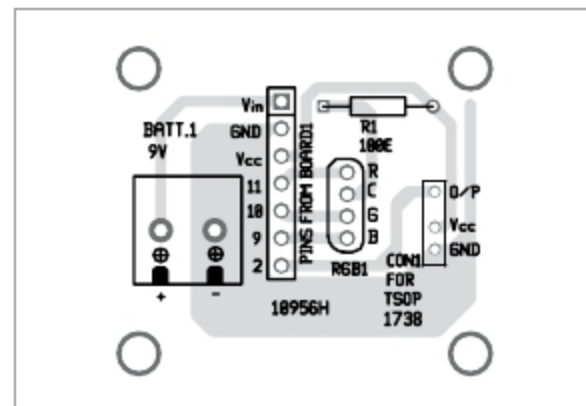


Fig. 6: Components layout for the PCB

are used to turn on and off RGB1, respectively. Message 'RGB bulb on' or 'RGB bulb off' is also displayed on Arduino's serial monitor.

Button 1 is used to increase the intensity of red LED and button 2 to decrease the intensity of red LED. Pressing these buttons increases or decreases the pulse width on digital output pin 9 from one to 100 per cent, to change the intensity of red LED. Message 'Red intensity is increased by XX%' or 'Red intensity is decreased by XX%' is displayed on the serial monitor.

Buttons 3 and 4 are used to increase and decrease green colour intensity, respectively. These buttons increase or decrease pulse width on digital output pin 10 of Arduino. Similarly, buttons 5 and 6 are used to increase and decrease blue colour intensity, respectively. These two buttons increase or decrease the pulse width on digital output pin 11 of Arduino.

Here, a 9V battery is used as power supply for Arduino Uno R3. **EFY**

EFY Note

The source code of this project is available for free download at source.efymag.com



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